

PrevPredMap: Exploring Temporal Modeling with Previous Predictions for Online Vectorized HD Map Construction

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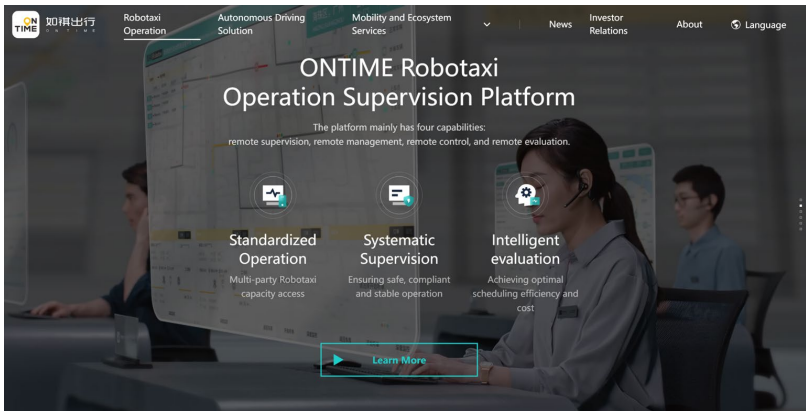
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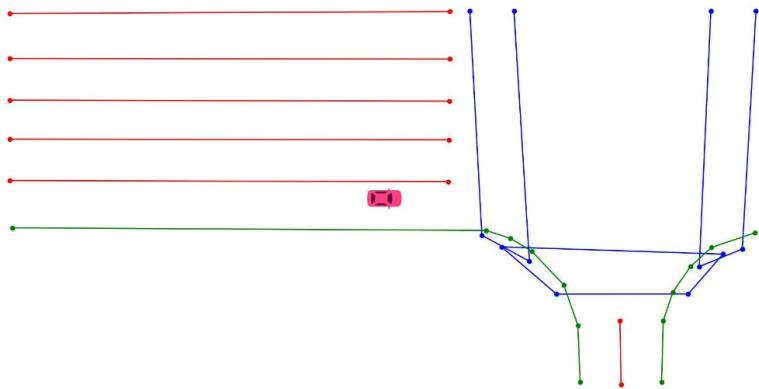
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- Problem/Objective
 - (Temporal) HD Map Construction
- Contribution/Key Idea
 - Novel Temporal method
 - Enhanced Single-frame baseline
 - Temporal / single 모두 잘 작동
 - SOTA



- **HD map construction**

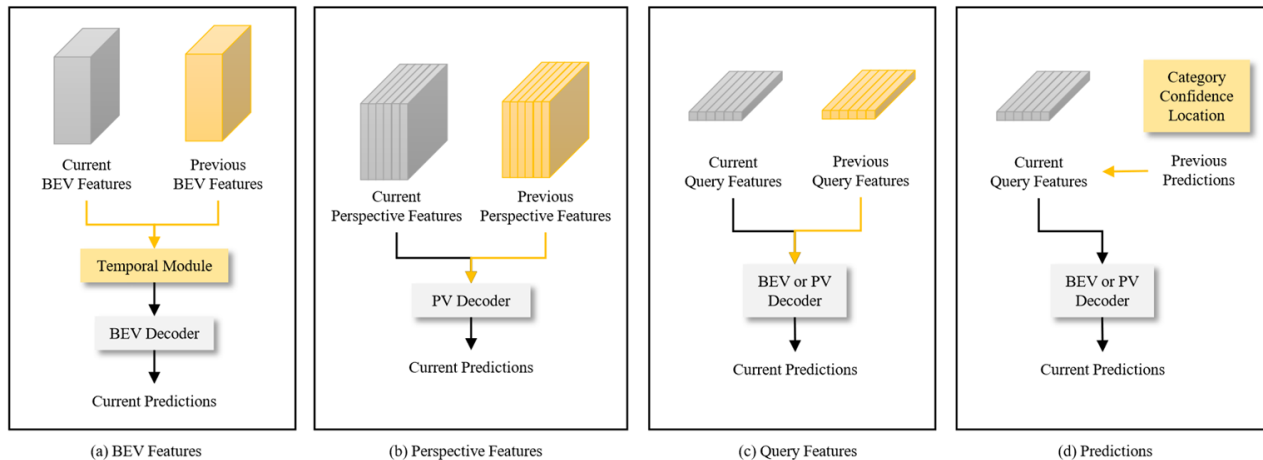


HD map construction [1]

- **Map element**를 point로 구성된 polyline으로 표현
- Loss → classification / point 2가지

[1] <https://github.com/Tsinghua-MARS-Lab/Online-HD-Map-Construction-CVPR2023#task>

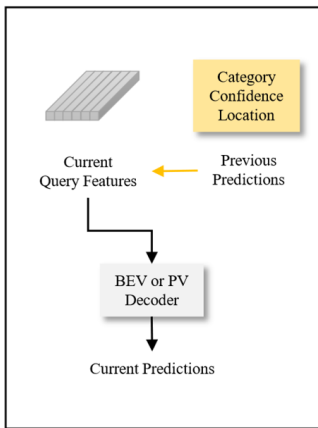
● Previous research



Various temporal representations

- **Temporal + BEV network** 이전 연구들은 크게 3가지
 - BEV / PV / Query - level “fusion or concat”
- Prediction 결과를 직접 이용 study 거의 x

- Two potential Advantages of leveraging Previous Prediction



(d) Predictions

- Prediction 결과** → 이미 semantic 의미를 가진 **high-level abstraction** (>feature)
- Prediction 결과** → vectorized된 표현 → 모델 구조적으로 map priors와 쉽게 통합 가능

→ 정확성 + 확장성 appeal

● Method

○ Group-wise one-to-many branch

- **MapTRv2**: 모델이 빠르게 수렴하도록 하기 위해, 학습 중 일부 쿼리들을 **GT와 1:N 매칭**해서 추가적인 학습 **supervision**을 제공 (DETR 류에서 많이 사용)

○ 단점:

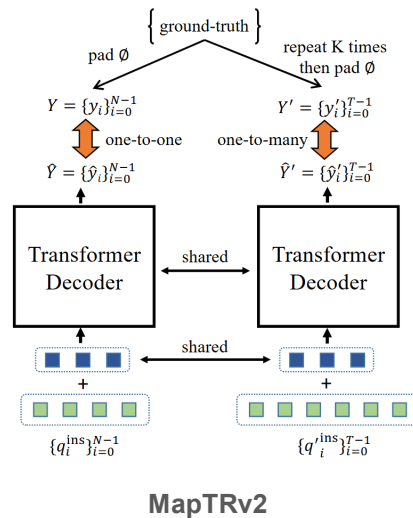
- 쿼리 수가 많아지면 → self-attention의 **memory 사용량이 $O(N^2)$** 로 급증
- GT 하나에 여러 query가 대응되므로 memory up..

- 본 논문: Group-wise one-to-many branch

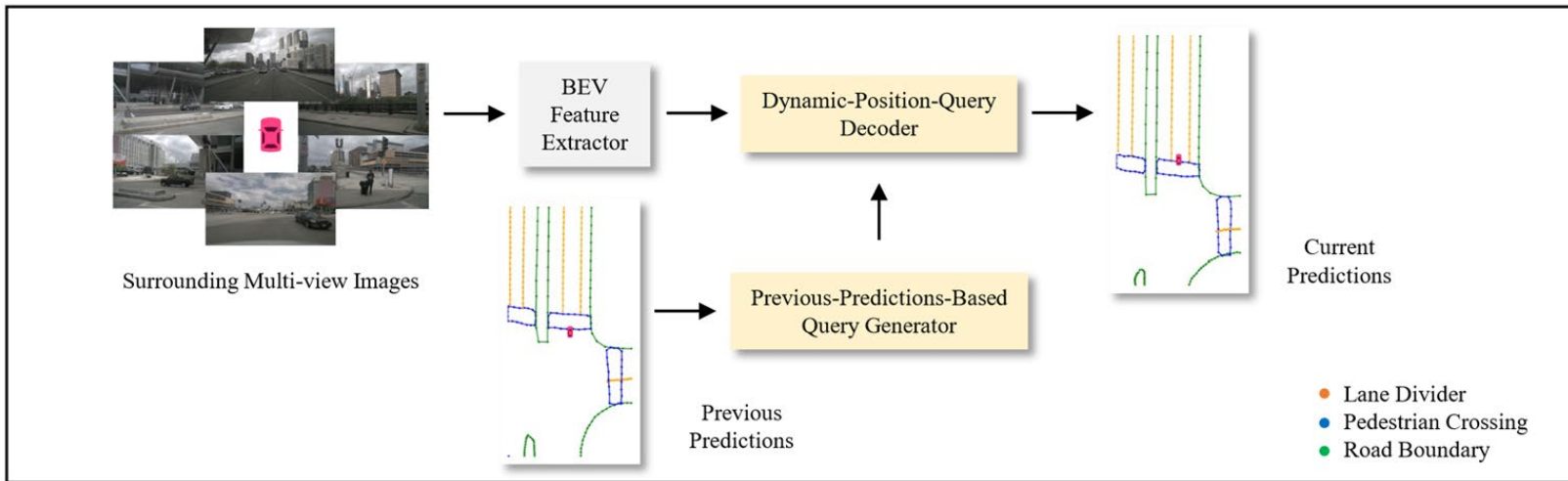
- 전체 쿼리를 여러 **그룹(group)**으로 나눠서 **병렬로 one-to-many matching**을 수행
- Self-attention 연산이 group 단위로 분할되므로 메모리 사용량이 줄어듦
 - $O((N/G)^2 \times G) = O(N^2/G)$ 수준으로 효율화 가능

- 이렇게 해서 쿼리 수를 증가시켜도 메모리 부담 없이 **strong한 single-frame baseline**을 만들 수 있었다~

5.3 Auxiliary One-to-Many Set Prediction Loss



● Method



(a) Pipeline Overview

- **BEV Feature Extractor : 2D CNN + BEV transform**
- **Previous-Predictions-Based Query Generator**
- **Dynamic-Position-Query Decoder**

● Method

● Previous-Predictions-Based Query Generator

- **Dual-Mode** 학습 전략
 - Single Frame mode / Temporal mode (확률로 랜덤 select)
- Content(무엇을) / Position(어디서) Query 두개를 생성
 - MapTR 스타일 계승 (j-th point of the i-th map element)
- **Non-Previous-Predictions-Based Query (Single-frame mode)**

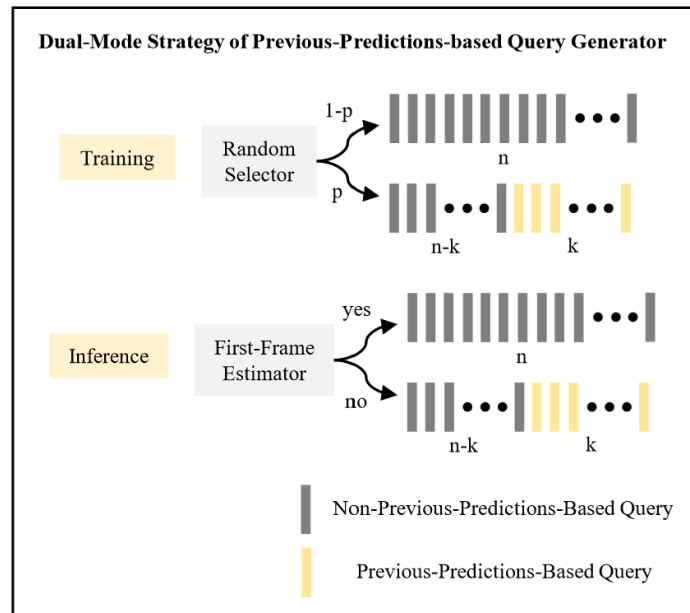
$$q_{ij}^{con} = q_i^{con-ins} + q_j^{con-pt},$$

$$q_{ij}^{pos} = q_i^{pos-ins} + q_j^{pos-pt},$$



Previous x (1-p) :

$$q_{ij}^{pos} = q_i^{pos-ins} + PE(q_j^{loc-pt}),$$



(b) Previous-Predictions-Based Query Generator

● Method

● Previous-Predictions-Based Query Generator

○ Previous-Predictions-Based Query (Temporal mode)

- Category / Confidence / location을 encoding 하여 query로

$$q_{ij}^{con} = q_i^{con-ins} + q_j^{con-pt},$$

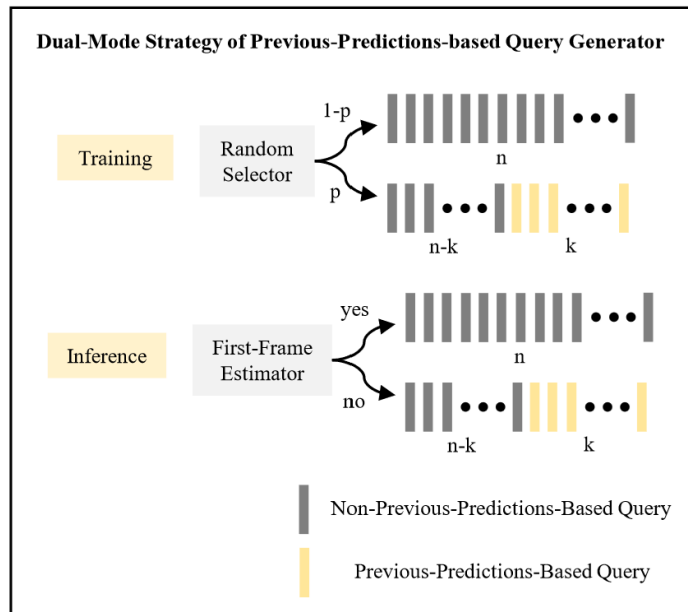
$$q_{ij}^{pos} = q_i^{pos-ins} + q_j^{pos-pt},$$



$$q_{ij}^{con} = Linear(q_i^{cate}) + Linear(q_i^{conf}) + q_j^{con-pt},$$

Previous o (p) :

$$q_{ij}^{pos} = PE(q_{ij}^{loc}),$$



(b) Previous-Predictions-Based Query Generator

● Method

● Previous-Predictions-Based Query Generator

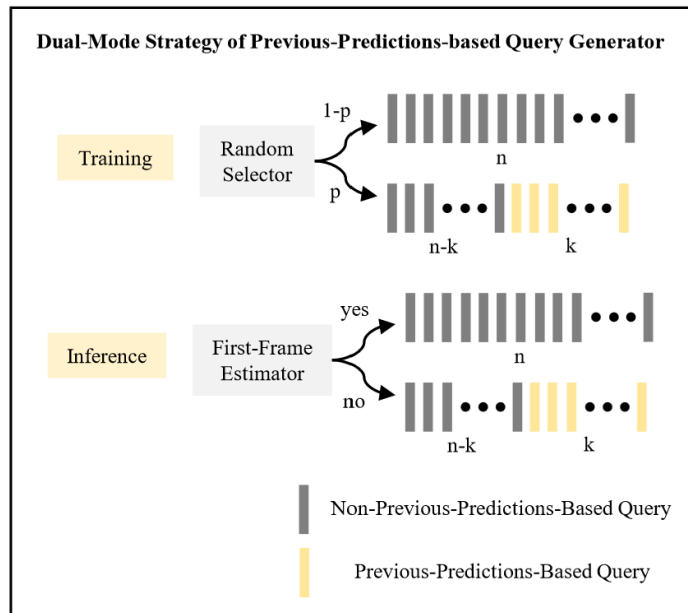
- Training에서는 temporal / single-frame 혼합하여 사용
 - decoder가 받는 위치 쿼리의 분포를 통일하기 위해
 - 위치 표현 방식을 통일 하기 위해 PE를 사용

Single-frame (1-p) :

$$q_{ij}^{pos} = q_i^{pos-ins} + PE(q_j^{loc-pt}),$$

Temporal (p) :

$$q_{ij}^{pos} = PE(q_{ij}^{loc}),$$



(b) Previous-Predictions-Based Query Generator

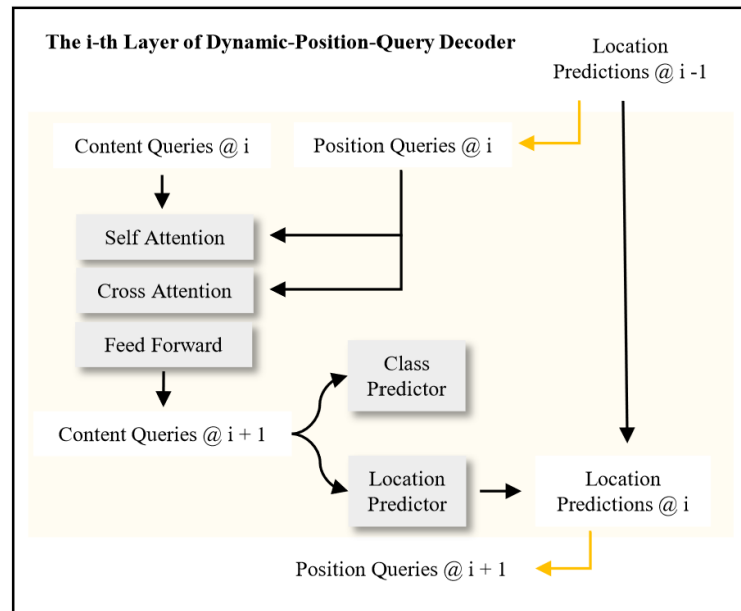
Method

Dynamic-Position-Query Decoder

- Self-Attention / Cross-Attention / Feed Forward / 위치 업데이트
- 기존 모델들 decoder의 모든 layer에서 동일한 position query 사용
 - query가 바라보는 spatial attention 위치는 고정
- 그래서
 - PrevPredMap은 layer마다 위치 정보를 갱신
 - 초기 layer ($n=0$) & $n>0$ 에서 다르게

즉 이전 layer의 위치 예측 결과를 기반으로 다음 position

$$q_{ij}^{pos}@n = \begin{cases} q_{ij}^{pos} & \text{if } n = 0 \\ PE(q_{ij}^{loc}@n) & \text{if } n > 0 \end{cases},$$



(c) Dynamic-Position-Query Decoder

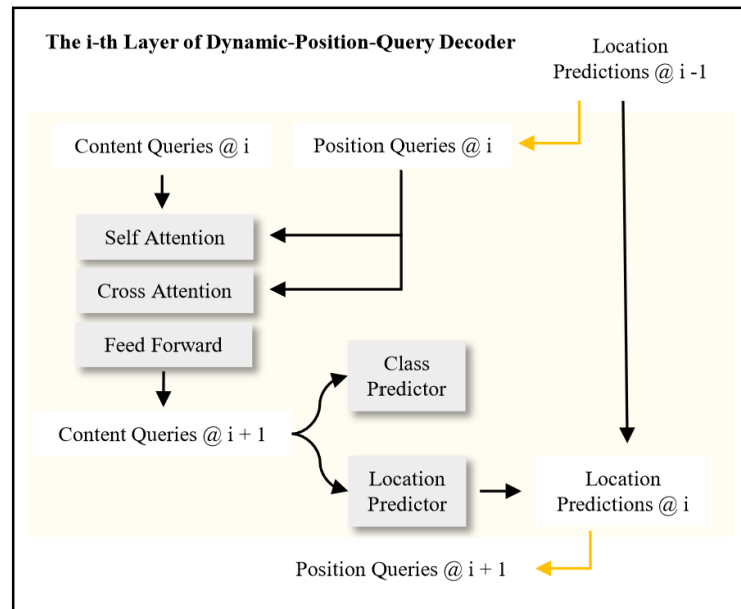
Method

Dynamic-Position-Query Decoder

- Self-Attention / Cross-Attention / Feed Forward / 위치 업데이트
- 기존 모델들 decoder의 모든 layer에서 동일한 position query 사용
 - query가 바라보는 spatial attention 위치는 고정
- 그래서
 - PrevPredMap은 layer마다 위치 정보를 갱신
 - 초기 layer ($n=0$) & $n>0$ 에서 다르게
- 즉, 이전 decoder layer의 위치 예측 결과를 기반으로 다음 position query를 업데이트

- 마지막 layer에서 point-wise 위치 / category / confidence 예측

$$q_{ij}^{pos}@n = \begin{cases} q_{ij}^{pos} & \text{if } n = 0 \\ PE(q_{ij}^{loc}@n) & \text{if } n > 0 \end{cases},$$



(c) Dynamic-Position-Query Decoder

• Experiment

Table 1. Comparison with SOTA methods on nuScenes. All backbones utilized are ResNet50. "Temp." signifies the utilization of temporal information. The * indicates that MapTRv2 has been re-implemented with the number of instance queries set to 100. The ★ denotes that FPS measurements were conducted on the same machine with NVIDIA RTX A6000 for fair comparison.

Method	Epoch	Temp.	AP_{div}	AP_{ped}	AP_{bou}	mAP	FPS
MapTR ICLR23 [18]	24	no	51.5	46.3	53.1	50.3	15.1
MapTRv2* arxiv23 [19]	24	no	62.8	62.0	65.4	63.4	16.8★
StreamMapNet WACV24 [40]	30	yes	66.3	61.7	62.1	63.4	14.9★
SQD-MapNet arxiv24 [32]	24	yes	66.6	63.6	64.8	65.0	14.9★
PrevPredMap (Ours)	24	yes	66.9	64.5	67.6	66.3	15.7★
MGMap CVPR24 [23]	24	no	65.0	61.8	67.5	64.8	11.6
HIMap CVPR24 [43]	30	no	68.4	62.6	69.1	66.7	11.4
MapQR ECCV24 [26]	24	no	68.0	63.4	67.7	66.4	14.7★
PrevPredMap++ (Ours)	24	yes	68.7	66.0	68.3	67.6	13.9★
MapTRICLR23 [18]	110	no	59.8	56.2	60.1	58.7	15.1
MapTRv2* arxiv23 [19]	110	no	68.8	68.0	71.0	69.2	16.8★
PrevPredMap (Ours)	110	yes	70.0	71.2	72.8	71.3	15.7★

• Experiment

Table 2. Comparison with SOTA methods on Argoverse2. All backbones utilized are ResNet50. "Dim." refers to the dimension of the predicted coordinates for map elements. "Freq." denotes the sampling frequency applied during training. "Temp." signifies the utilization of temporal information. The * indicates that MapTRv2 has been re-implemented with the number of instance queries set to 100.

Method	Dim.	Epoch	Freq.	Temp.	AP_{div}	AP_{ped}	AP_{bou}	mAP
MapTRv2* arxiv23 [19]	2d	30	2Hz	no	70.9	61.7	65.8	66.1
StreamMapNet WACV24 [40]	2d	30	2Hz	yes	62.0	59.5	63.0	61.5
SQD-MapNet arxiv24 [32]	2d	30	2Hz	yes	64.9	60.2	64.9	63.3
PrevPredMap(Ours)	2d	30	2Hz	yes	72.0	64.2	68.3	68.2
MapTRv2* arxiv23 [19]	2d	6	10Hz	no	73.3	65.3	69.1	69.2
HIMap CVPR24 [43]	2d	6	10Hz	no	69.5	69.0	70.3	69.4
ADMapv2 ECCV24 [10]	2d	6	10Hz	no	72.4	64.5	68.9	68.7
PrevPredMap(Ours)	2d	6	10Hz	yes	74.8	65.7	70.9	70.5
MapTRv2* arxiv23 [19]	3d	6	10Hz	no	71.7	62.2	66.7	66.9
PrevPredMap(Ours)	3d	6	10Hz	yes	71.4	64.1	67.4	67.6

• Experiment

Table 3. Ablation of the proposed PrevPredMap. The experiments were performed on nuScenes, following a 24-epoch training schedule. FPS measurements are conducted on the same machine with NVIDIA RTX A6000.

Query Amount	Group-Wise One-to-Many	Previous-Predictions-based Query Generator	Dynamic-Position-Query Decoder	mAP	FPS
50				61.5	16.8
100				63.4	16.8
100	✓			64.8	16.8
100	✓	✓		65.4	16.1
100	✓		✓	64.9	16.0
100	✓	✓	✓	66.3	15.7

• Experiment

Table 4. Performance comparison of the single-frame and temporal modes for PrevPredMap on nuScenes, following a 24-epoch training schedule. FPS measurements are conducted on the same machine with NVIDIA RTX A6000.

Mode	mAP	AP_{div}	AP_{ped}	AP_{bou}	FPS
Single-Frame	65.7	66.0	63.9	67.3	16.1
Temporal	66.3	66.9	64.5	67.6	15.7

Table 5. Probability of the previous-predictions-based query generator. All experiments were conducted on nuScenes, following a 24-epoch training schedule.

Probability	mAP	AP_{div}	AP_{ped}	AP_{bou}
0.4	66.2	65.8	65.4	67.3
0.5	66.3	66.9	64.5	67.6
0.6	65.7	65.8	63.8	67.5

- Experiment

Table 6. The number of previous-predictions-based queries. All experiments were conducted on nuScenes, following a 24-epoch training schedule.

k	mAP	AP_{div}	AP_{ped}	AP_{bou}
5	66.1	65.5	65.0	67.5
8	66.3	65.8	64.9	68.2
10	66.3	66.9	64.5	67.6
12	66.5	66.9	64.5	68.0
15	65.7	65.3	64.8	66.9

- 12가 best but, 10개만 사용
 - 한 프레임에는 map element 10개 이하

- Experiment

